

Projectile motion.

Question-1: A projectile is fired from ground level with an initial speed of 50 m/s at an angle 53° to the horizontal.

i) calculate the maximum height, flight time, horizontal range

Answer: Given initial velocity $u_0 = 50 \text{ m/s}$, angle $\theta = 53^\circ$

$$\text{The equation of maximum height } H = \frac{u_0^2 \sin^2 \theta}{2g} = \frac{50^2 \sin^2 53}{2 \times 9.8} = 81.35 \text{ m}$$

$$\text{Total time } T = \frac{2u_0 \sin \theta}{g} = \frac{2 \times 50 \times \sin 53}{9.8} = 8.15 \text{ sec}$$

$$\text{Range } R = \frac{u_0^2 \sin 2\theta}{g} = 245.2 \text{ m}$$

ii) Find position (x, y) and speed at $t = 3 \text{ sec}$

Answer: The horizontal distance

$$x = u_0 \cos \theta \times t = 50 \cos 35^\circ \times 3 = 90.27 \text{ m}$$

The vertical distance

$$y = u_0 \sin \theta t - \frac{1}{2} g t^2 = 50 \sin 35^\circ \times 3 - \frac{1}{2} \times 9.8 \times (3)^2 = 75.69 \text{ m}$$

$$\text{Position } (x, y) = (90.27, 75.69)$$

$$\text{The horizontal velocity component } V_x = u_0 \cos \theta = 50 \cos 35^\circ = 30.09 \text{ m/s}$$

$$\text{The vertical velocity component } V_y = u_0 \sin \theta - g t = 50 \sin 35^\circ - 9.8 \times 3 = 10.53 \text{ m/s}$$

$$\text{Total velocity } V = \sqrt{V_x^2 + V_y^2} = \sqrt{30.09^2 + 10.53^2} = 31.53 \text{ m/s}$$

iii) Angle the velocity vector at $t = 3 \text{ sec}$.

Answer: Since, $V_x = 30.09 \text{ m/s}$

$$V_y = 10.53 \text{ m/s}$$

The tangent is $\tan \theta = \frac{V_y}{V_x}$

$$\Rightarrow \theta = \tan^{-1} \left(\frac{V_y}{V_x} \right) = \tan^{-1} \left(\frac{10.53}{30.09} \right)$$

Question-2: A ball is thrown from the top of a building 70m high with an initial speed of 25 m/s at an angle of 30° above horizontal.

i) Time taken to reach the ground.

Answer: The equation of motion is given by,

$$y = V_0 \sin \theta t - \frac{1}{2} g t^2$$

$$\Rightarrow -70 = 25 \sin 30^\circ t - \frac{1}{2} g t^2$$

$$\Rightarrow g t^2 - 50 \sin 30^\circ t - 70 = 0$$

$$\therefore t = 5 \text{ sec.}$$

ii) The horizontal range from the base of the building

Answer: Horizontal displacement is given by,

$$x = V_0 \cos \theta \times t$$

$$= 25 \cos 30^\circ \times 5$$

ii) Resultant velocity (magnitude and direction) just before impact.

Answer: The horizontal velocity component

$$V_x = V_0 \cos \theta$$

The vertical component of velocity

$$V_y = V_0 \sin \theta - g t$$

Resultant velocity

$$V = \sqrt{V_x^2 + V_y^2}$$

angle

$$\theta = \tan^{-1} \left(\frac{V_y}{V_x} \right)$$

iii) Angle the velocity vector at $t = 3 \text{ sec}$.

Answer: Since, $V_x = 30.09 \text{ m/s}$

$$V_y = 10.53 \text{ m/s}$$

The tangent is $\tan \theta = \frac{V_y}{V_x}$

$$\Rightarrow \theta = \tan^{-1} \left(\frac{V_y}{V_x} \right) = \tan^{-1} \left(\frac{10.53}{30.09} \right)$$

Question-2: A ball is thrown from the top of a building 70m high with an initial speed of 25 m/s at an angle of 30° above horizontal.

i) Time taken to reach the ground.

Answer: The equation of motion is given by,

$$y = V_0 \sin \theta t - \frac{1}{2} g t^2$$

$$\Rightarrow -70 = 25 \sin 30^\circ t - \frac{1}{2} g t^2$$

$$\Rightarrow g t^2 - 50 \sin 30^\circ t - 70 = 0$$

$$\therefore t = 5 \text{ sec.}$$

ii) The horizontal range from the base of the building

Answer: Horizontal displacement is given by,

$$x = V_0 \cos \theta \times t$$

$$= 25 \cos 30^\circ \times 5$$

ii) Resultant velocity (magnitude and direction) just before impact.

Answer: The horizontal velocity component

$$V_x = V_0 \cos \theta$$

The vertical component of velocity

$$V_y = V_0 \sin \theta - g t$$

Resultant velocity

$$V = \sqrt{V_x^2 + V_y^2}$$

angle

$$\theta = \tan^{-1} \left(\frac{V_y}{V_x} \right)$$

Question-1:

3) A spring mass system $m=200g$, $k=50N/m$, pulled $A=5cm$ from equilibrium, released from rest (i) Express the equation of motion and calculate time period.

Answer: Given mass $m=200g=0.2kg$, Amplitude $A=5cm=0.05m$

Spring constant $k=50N/m$

The angular velocity $\omega = \sqrt{k/m} = \sqrt{50/0.2} = 15.81 \text{ rad/s}$

Since the ideal equation of motion,

$$x = A \cos(\omega t) \\ = 0.05 \cos(15.81 t)$$

ii) compute the velocity at displacement $x=3cm$

Answer: The velocity is given by,

$$v = \omega \sqrt{A^2 - x^2} = 15.81 \sqrt{(0.05)^2 - (0.03)^2}$$

$$\therefore v = 0.632 \text{ m/s}$$

iii) find the total mechanical energy of the system.

Answer: The total mechanical Energy equation,

$$E = \frac{1}{2} k A^2 = \frac{1}{2} \times 50 \times (0.05)^2 = 0.0625 J$$

Question-2: A particle execute SHM represented by equation $y = 15 \sin(10t - \pi/4)$ [cm]

i) Angular frequency, frequency and time period.

from ideal equation of motion,

$$y = A \sin(\omega t + \phi)$$

Angular frequency $\omega = 10 \text{ rad/s}$, frequency $f = \frac{\omega}{2\pi} = 1.59 \text{ Hz}$, Time period

$$T = 1/f = 0.628 \text{ sec}$$

ii) maximum velocity and acceleration.

Answer: The velocity equation is $v = \omega \sqrt{A^2 - x^2}$

maximum velocity occurs at $x=0$, so,

$$v = \omega A = 15 \times 10 = 150 \text{ cm/s}$$

The maximum acceleration $a = \omega^2 A$

$$= 100 \times 15 = 1500 \text{ cm/s}^2$$

iii) displacement, velocity and acceleration at $t=2\text{sec}$

Answer: The equation of displacement

$$y = 15 \sin(10t - \pi/4)$$

$$\text{at } t=2 \text{ sec} \\ y = 15 \sin[10 \times 2 - \pi/4] = 14.028 \text{ m}$$

The velocity equation

$$v = A\omega \cos(\omega t - \pi/4)$$

$$\text{at } t=2 \text{ sec, } v = 15 \times 10 \cos[10 \times 2 - \pi/4] = -53.3 \text{ cm/s}$$

The acceleration equation,

$$a = -A\omega^2 \sin(\omega t - \pi/4)$$

$$\text{at } t=2 \text{ sec,}$$

$$a = -15 \times 10^2 \sin[10 \times 2 - \pi/4]$$

$$= -1402 \text{ cm/s}^2$$

iv) Potential and kinetic energy at $x=8 \text{ cm}$. if mass of particle

Answer: The potential energy is

$$E_p = \frac{1}{2} m \omega^2 x^2 = \frac{1}{2} k x^2$$

$$= \frac{1}{2} \times 0.5 \times (0.1)^2 \times (0.08)^2$$

$$\begin{aligned} \text{Kinetic energy} \\ E_k &= E - E_p \\ &= \frac{1}{2} k A^2 - E_p \\ &= \frac{1}{2} m \omega^2 A^2 - E_p \end{aligned}$$

is $m = 0.5 \text{ kg}$

$$\omega = 10 \text{ cm}$$

$$= 0.1 \text{ m}$$

$$x = 8 \text{ cm} = 0.08 \text{ m}$$

Rotational mechanics

Question: A particle moves in a circle of radius 8m. Its linear speed varies as $v = 3t^2 + 4t$

i) calculate centripetal and tangential acceleration at $t = 3\text{sec}$

Answer: Velocity at $t = 3\text{sec}$,

$$v = 3 \times 3^2 + 4 \times 3 = 39 \text{ m/s}$$

$$\text{The centripetal acceleration } a_c = \frac{v^2}{r} = \frac{39^2}{8} = 190 \text{ m/s}^2$$

The tangential acceleration

$$a_t = \frac{dv}{dt} = 6t + 4 = 6 \times 3 + 4 = 22 \text{ m/s}^2$$

ii) magnitude of resultant acceleration

$$a_r = \sqrt{a_c^2 + a_t^2} = \sqrt{190^2 + 22^2} = 191.4 \text{ m/s}^2$$

ii) Angle between resultant acceleration and the radius vector at $t = 3\text{sec}$

Answer: The angle between a_r and r is

$$\tan \theta = a_t / a_c$$

$$\Rightarrow \theta = \tan^{-1} \left(\frac{a_t}{a_c} \right) = \tan^{-1} \left(\frac{22}{190} \right)$$

$$\therefore \theta = 6.6^\circ$$

Question: A solid disk ($m = 8 \text{ kg}$, $R = 0.4 \text{ m}$) rotates at 130 rpm.

A braking torque of 2 N.m is applied.

i) calculate moment of inertia | ii) calculate deceleration produced.

Answer: The moment of inertia

$$\text{for disk is } I = \frac{1}{2} m R^2$$

$$= \frac{1}{2} \times 8 \times 0.4^2$$

Answer: The torque is given by,

$$\tau = I \alpha$$

$$\Rightarrow \alpha = \frac{\tau}{I} = \frac{2}{\dots}$$

iii) Revolutions before coming to rest.

Answer: The initial angular velocity $\omega_0 = 120 \text{ rpm}$

$$= 120 \times \frac{2\pi}{60} = 4\pi \\ = 12.57 \text{ rad/s}$$

$$\text{So, } \omega^2 = \omega_0^2 - 2\alpha\theta$$

$$\Rightarrow \theta = \frac{\omega^2 - \omega_0^2}{2\alpha} = \frac{0^2 - (12.57)^2}{2 \times (\text{num ii})}$$

Question: A block with a mass of 20 kg rests on earthbound plane elevated 35° off the horizontal. If the coefficient of kinetic friction for this surface is 0.54. What is the acceleration of the block.

Answer: The resultant equation for force is

$$\sum F = ma$$

$$\Rightarrow mg \sin \theta - \mu mg \cos \theta = ma$$

$$\Rightarrow a = g \sin \theta - \mu g \cos \theta$$

$$= 9.8 \sin 35^\circ - 0.54 \times 9.8 \times \cos 35^\circ$$

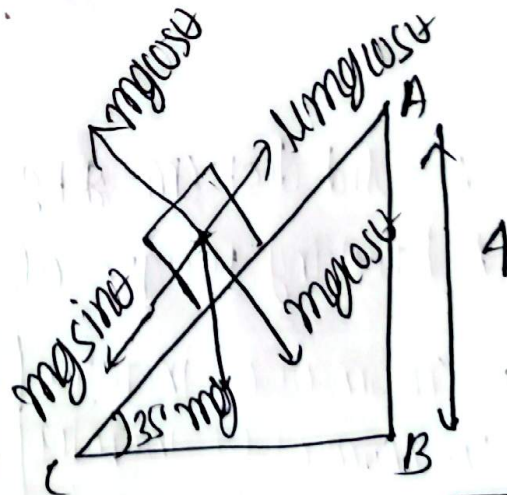
$$= 1.3 \text{ m/s}^2$$

ii) Time to reach the ground.

The distance AC is

$$\sin 35^\circ = \frac{4}{AC}$$

$$\Rightarrow AC = \frac{4}{\sin 35^\circ} = S = 5 \text{ m (assume)}$$



$$S = ut + \frac{1}{2}at^2 = \frac{1}{2}at^2 \Rightarrow 5 = \frac{1}{2} \times 1.3 \times t^2 \Rightarrow t = \sqrt{\frac{10}{1.3}} \approx 2.2 \text{ (assume)}$$